

Reporting Aggregated Data Using the Group Functions

1. Find the highest, lowest, sum, and average salary of all employees. Label the columns.

```
SQL> SELECT MAX(SAL) AS HIGHEST_SALARY, MIN(SAL) AS LOWEST_SALARY, SUM(SAL) AS TOTAL_SALARY, ROUND(AVG(SAL),2) AS AVERAGE_SALARY FROM EMP;
```

HIGHEST_SALARY	LOWEST_SALARY	TOTAL_SALARY	AVERAGE_SALARY
5000	800	29025	2073.21

2. Maximum, Minimum, Sum, and Average, respectively. Round your results to the nearest whole number.

```
SQL> SELECT ROUND(MAX(SAL)) AS Maximum, ROUND(MIN(SAL)) AS Minimum, ROUND(SUM(SAL)) AS Sum, ROUND(AVG(SAL)) AS Average FROM EMP;
```

MAXIMUM	MINIMUM	SUM	AVERAGE
5000	800	29025	2073

3. Modify the above query to display the minimum, maximum, sum, and average salary for each job type.

```
SQL> SELECT JOB, ROUND(MIN(SAL)) AS Minimum, ROUND(MAX(SAL)) AS Maximum, ROUND(SUM(SAL)) AS Sum, ROUND(AVG(SAL)) AS Average FROM EMP GROUP BY JOB;
```

JOB	MINIMUM	MAXIMUM	SUM	AVERAGE
PRESIDENT	5000	5000	5000	5000
MANAGER	2450	2975	8275	2758
ANALYST	3000	3000	6000	3000
CLERK	800	1300	4150	1038
SALESMAN	1250	1600	5600	1400

4. Write a query to display the number of people with the same job.

```
SQL> SELECT JOB, COUNT(*) AS TOTAL_EMPLOYEES FROM EMP GROUP BY JOB;
```

JOB	TOTAL_EMPLOYEES
PRESIDENT	1
MANAGER	3
ANALYST	2
CLERK	4
SALESMAN	4

5. Determine the number of managers without listing them. Label the column Number of Managers.

```
SQL> SELECT COUNT(*) AS "Number of Managers" FROM EMP WHERE JOB='MANAGER';
```

Number of Managers
3

6. Find the difference between the highest and lowest salaries. Label the column DIFFERENCE.

```
SQL> SELECT MAX(SAL)-MIN(SAL) AS DIFFERENCE FROM EMP;
```

```
DIFFERENCE
-----
4200
```

7. Create a report to display the manager number and the salary of the lowest-paid employee for that manager. Exclude anyone whose manager is not known. Exclude any groups where the minimum salary is \$2000 or less. Sort the output in descending order of salary.

```
SQL> SELECT MGR, MIN(SAL) AS LOWEST_SALARY FROM EMP WHERE MGR IS NOT NULL GROUP BY MGR HAVING MIN(SAL)>2000 ORDER BY LOWEST_SALARY DESC;
```

```
MGR LOWEST_SALARY
-----
7566          3000
7839          2450
```

8. Create a query to display the total number of employees and, of that total, the number of employees hired in 1980, 1981, and 1982. Create appropriate column headings.

```
SQL> SELECT COUNT(*) AS "Total Employees", COUNT(CASE WHEN TO_CHAR(HIREDATE,'YYYY')='1980' THEN 1 END) AS "1980", COUNT(CASE WHEN TO_CHAR(HIREDATE,'YYYY')='1981' THEN 1 END) AS "1981", COUNT(CASE WHEN TO_CHAR(HIREDATE,'YYYY')='1982' THEN 1 END) AS "1982" FROM EMP;
```

```
Total Employees      1980      1981      1982
-----
14              1          10          1
```